



Cyber Defense (420-D03-AB) · Class 11 · Wednesday, September 23, 2026

Network Device Security and Secure Communication Protocols

Protecting the equipment that carries NorthMaple's traffic, and the channels used to manage it.

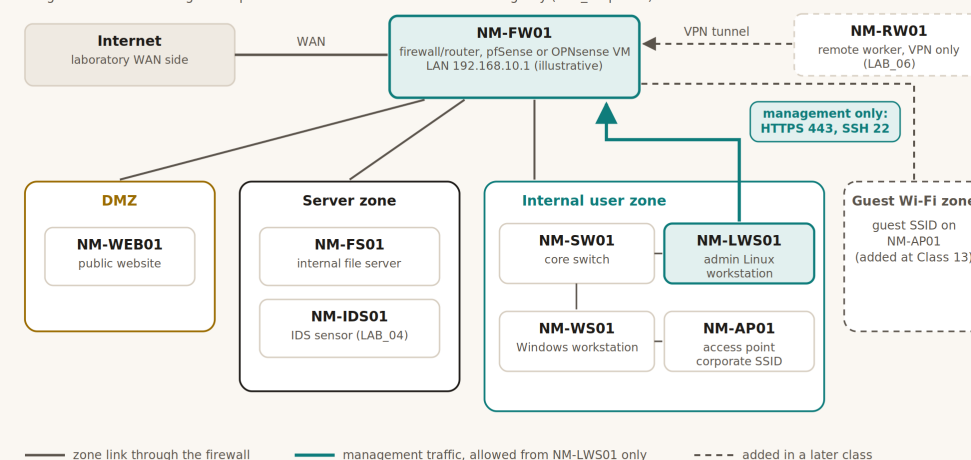
6:00 pm **EDT** (Eastern Daylight Time) · short break at 7:30 pm · then LAB_05, part 1 of 2

Cybersecurity **AEC** (Attestation d'études collégiales) · LEA.D8 · Competency EZ03, To apply access control measures



NorthMaple Accounting Inc. - laboratory topology for Class 11

Tonight: secure the management plane of NM-FW01 and start the device registry (LAB_05 part 1)



Host names follow the Chapter 1 diagram convention; addresses are illustrative - your LAB_05 brief of record wins.

Abbreviations: VM (Virtual Machine); LAN (Local Area Network); WAN (Wide Area Network); DMZ (Demilitarized Zone); IDS (Intrusion Detection System); SSID (Service Set Identifier); VPN (Virtual Private Network); HTTPS (Hypertext Transfer Protocol Secure); SSH (Secure Shell).

The NorthMaple Accounting Inc. laboratory (fictional company)

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Quick recall: what you have already built



Class 2: zones

Internet, **DMZ** (Demilitarized Zone), internal user zone and server zone, plus a **zoning table** listing every allowed flow with a one-sentence justification.



Classes 5 and 6: hosts

Hardened Windows and Linux machines, each with host firewall rules matched to its role.



Classes 7 and 8: detection

An **IDS** (Intrusion Detection System) sensor that watches the traffic and raises alerts.



All of these measures depend on equipment we have not secured yet: the firewall/router, the switch and the access point. An attacker who logs in to the firewall as its administrator can rewrite every rule of the zoning table.

Network device: equipment whose main job is to move or filter traffic between other systems (routers, switches, firewalls, wireless access points).

Data in transit: data that is moving across a network. Data at rest is stored on a disk or another storage medium.

Tonight's plan: four lessons, then LAB_05



1

Network device security

16 min

Routers, switches, firewalls and wireless equipment: default credentials, management access, firmware.



2

Secure communication protocols

18 min

Encrypted channels for remote administration, web management and tunnels between sites.



3

Update and patch management

9 min

Advisories, priorities, a written policy and a safe update sequence.



4

Device registry and configuration plan

9 min

What the company owns, in what state, and how changes are controlled.



LAB

LAB_05, part 1 of 2

after the 7:30 pm break

Deploy pfSense or OPNsense as a **VM** (Virtual Machine), secure its management access, start the device registry.



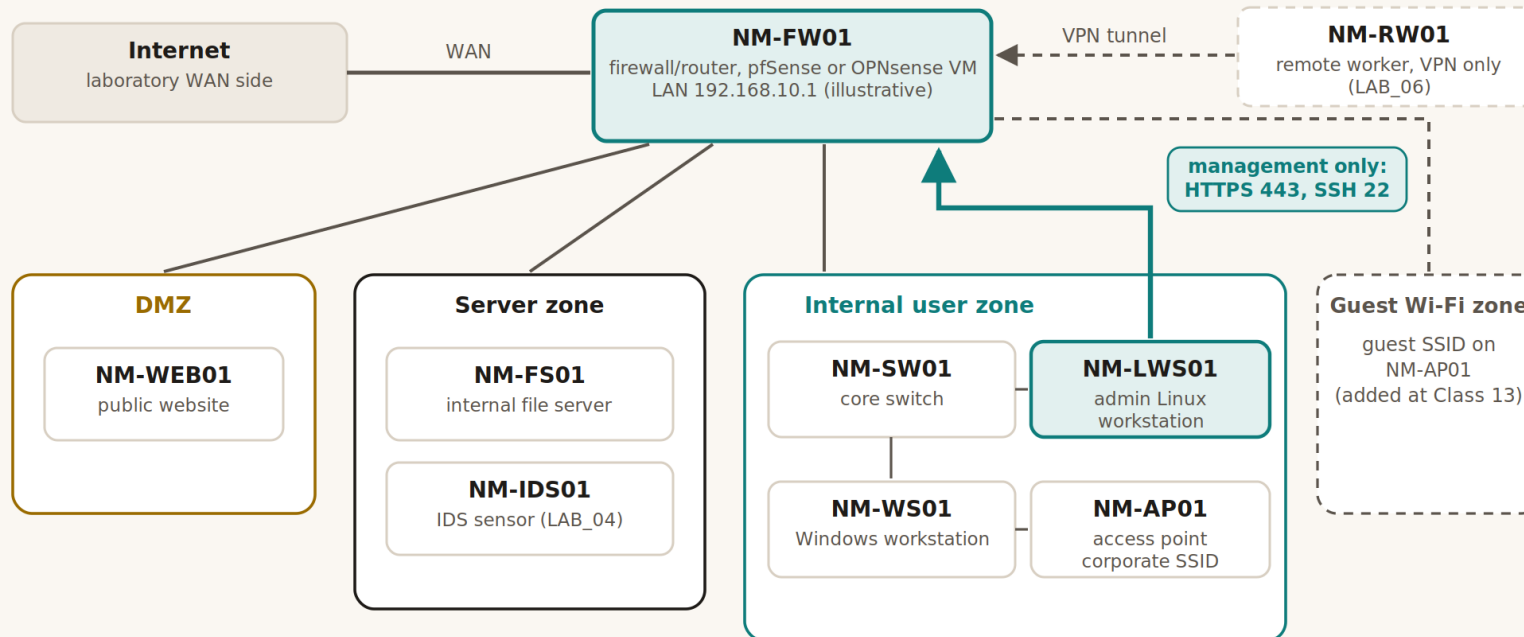
Every lesson prepares one task of LAB_05. After the break you apply them in this order on your own laboratory firewall.

The NorthMaple laboratory we secure tonight

Figure 0

NorthMaple Accounting Inc. - laboratory topology for Class 11

Tonight: secure the management plane of NM-FW01 and start the device registry (LAB_05 part 1)



Abbreviations: VM (Virtual Machine); LAN (Local Area Network); WAN (Wide Area Network); DMZ (Demilitarized Zone); IDS (Intrusion Detection System); SSID (Service Set Identifier); VPN (Virtual Private Network); HTTPS (Hypertext Transfer Protocol Secure); SSH (Secure Shell).



One idea for tonight

A network device handles two kinds of traffic:

- the traffic it forwards for the users;
- the traffic that configures and controls the device itself.

Whoever controls the second kind controls the first.

Dashed path in the figure: management traffic, allowed only from the administrator workstation NM-LWS01.



Four types of network device



Router

Connects different **IP** (Internet Protocol) networks. For each packet it reads the destination **IP** address, checks its **routing table** and sends the packet out of the matching interface.



Switch

Connects devices inside one **LAN** (Local Area Network). It learns which **MAC** (Media Access Control) address is reachable on which port and forwards each frame only to the port of the destination.



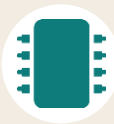
Firewall

Allows or blocks traffic between networks with an ordered list of rules. A **stateful firewall** also tracks active connections and accepts their replies.



Wireless access point

Connects Wi-Fi clients to the wired network and enforces the **SSID** (Service Set Identifier), the encryption mode and the passphrase.



Tonight's pfSense or OPNsense **VM** (Virtual Machine) is a router, a stateful firewall, a **DHCP** (Dynamic Host Configuration Protocol) server and a **DNS** (Domain Name System) resolver in one appliance. Network devices run an **embedded operating system**: only the manufacturer can update it, so your configuration and your updates are their main protection.

The three planes of a network device (1/2)

What each plane does



Data plane

Also called the **forwarding plane**.
Moves user traffic: an accountant's file transfer to the file server, a visitor's request to the company website.



Control plane

The processes that make the forwarding decisions: routing protocols that exchange paths with other routers, the process that builds the switch's address table.



Management plane

The services that let a person or a program configure and monitor the device: the web **GUI** (Graphical User Interface), the **CLI** (Command-Line Interface) over **SSH** (Secure Shell) or a console cable, **SNMP** (Simple Network Management Protocol) monitoring and **APIs** (Application Programming Interfaces).

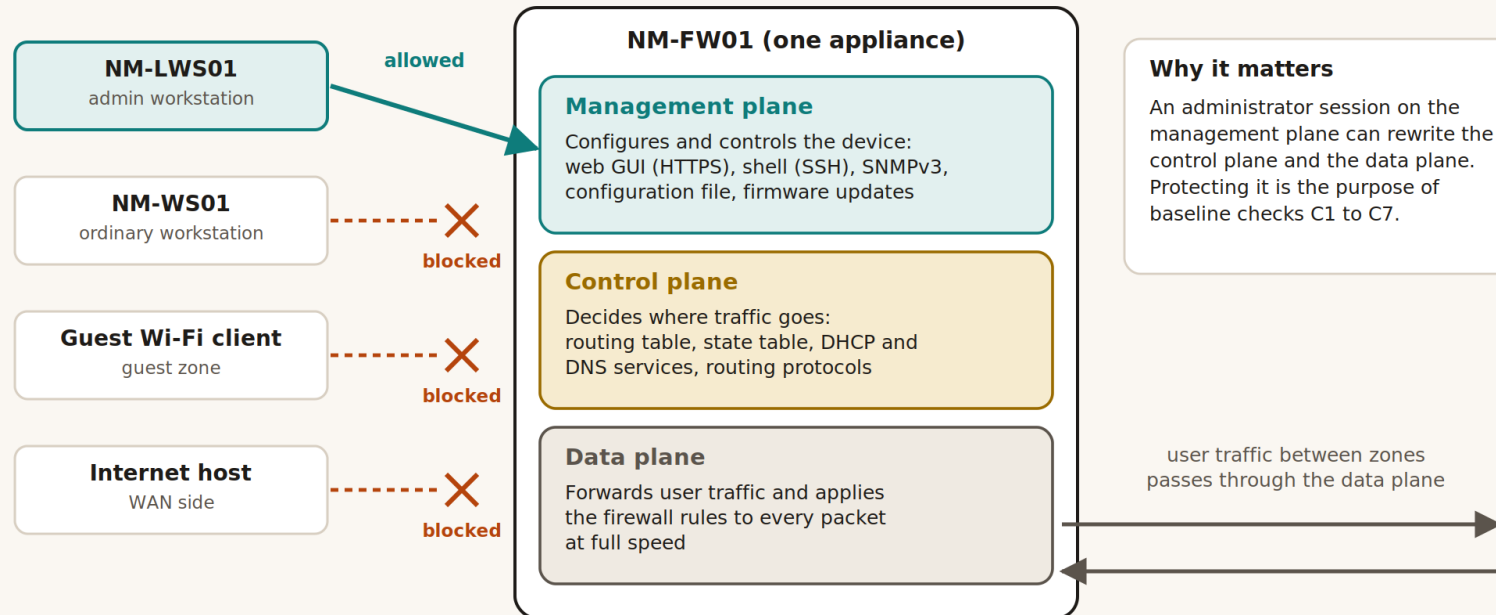


An administrator session on the management plane can rewrite the control plane and the data plane. That is why tonight starts with the management plane.

The three planes of a network device (2/2)

Who may reach the management plane (Figure 1)

Figure 1. The three planes of NM-FW01 and who may reach the management plane



Abbreviations: GUI (Graphical User Interface); HTTPS (Hypertext Transfer Protocol Secure); SSH (Secure Shell); SNMPv3 (Simple Network Management Protocol version 3); DHCP (Dynamic Host Configuration Protocol); DNS (Domain Name System); WAN (Wide Area Network).

Figure 1. Only NM-LWS01 may open a management session (**HTTPS** (Hypertext Transfer Protocol Secure) or **SSH** (Secure Shell)).



Why it matters

Attack surface: the complete set of points through which someone can try to interact with a system.

On a network device the management plane is its most valuable part: an administrator account can change every rule, disable logging and redirect traffic.

- **CIS** (Center for Internet Security) Safeguard 4.2: a documented secure configuration process for network infrastructure.
- **CIS Safeguard 4.6:** administrative interfaces only over secure protocols such as **SSH** and **HTTPS**, never Telnet or **HTTP** (Hypertext Transfer Protocol).

Default credentials (1/2)

The problem

- A **default credential** is a username and password set by the manufacturer before the device leaves the factory.
- Typical example: an account called admin or root, with a password printed in the manual, on a label or on the vendor's support website.
- The same values are used on thousands of units and are publicly documented.
- Result: they provide **no authentication**. Anyone who knows the model knows the password.
- MITRE **ATT&CK** (Adversarial Tactics, Techniques, and Common Knowledge), a public knowledge base of attacker techniques, lists the use of default accounts as technique **T1078.001**.

2016

The Mirai malware scanned the Internet for devices that answered on Telnet.

about 60

factory username and password pairs in its built-in list.

DDoS

Several hundred thousand infected cameras, video recorders and home routers launched some of the largest **DDoS** (Distributed Denial-of-Service) attacks of that time.



Default credentials (2/2)

Five controls, applied before the device joins any production network

- 1 Change the default password at the first login.** pfSense and OPNsense both ask you to replace it during setup.
- 2 Use a long, unique passphrase for each device** (several words): a password discovered on one device opens no other one (**CIS** (Center for Internet Security) Safeguard 5.2).
- 3 Give each administrator a named account.** Keep the built-in account for emergencies or disable it: the log shows who made each change.
- 4 Store device passwords in a password manager**, never in the registry, a spreadsheet or a screenshot. Course passwords are laboratory-only.
- 5 Add a second factor to the web GUI** (Graphical User Interface) **login** when the device supports it: OPNsense accepts **TOTP** (Time-based One-Time Password) codes from an authenticator application.



CIS Safeguard 4.7

"Manage Default Accounts on Enterprise Assets and Software". The five controls implement it.

Before



admin / factory password

After



named account +
passphrase + second factor

Securing management access (1/2)

Four questions: from where, and over which protocol

1 From where?



From where?

- **Management access restriction:** the management services accept connections only from approved sources.
- At NorthMaple the only approved source is the administrator's Linux workstation, NM-LWS01.
- Never from the **WAN** (Wide Area Network) interface (the Internet side) or from the guest Wi-Fi.
- A separate path for management is **out-of-band** management; the same networks as user data is **in-band**. NorthMaple: in-band, restricted to one source.

2 Over which protocol?



Over which protocol?

- Only encrypted protocols: **HTTPS** (Hypertext Transfer Protocol Secure) for the **web GUI** (Graphical User Interface), **SSH** (Secure Shell) for the command line.
- **HTTP** (Hypertext Transfer Protocol) and Telnet are disabled: they send the password in cleartext.
- **SNMPv1** (Simple Network Management Protocol version 1) and **SNMPv2c** (Simple Network Management Protocol version 2c) authenticate with a **community string** that also travels in cleartext. If monitoring is needed: **SNMPv3** (Simple Network Management Protocol version 3) with authentication and encryption.
- Lesson 2 explains how these protocols protect the session.

3 By whom?

4 With what record?

Securing management access (2/2)

By whom, and with what record

1 From where?

2 Over which protocol?

3 By whom?

4 With what record?



By whom?

- **Least privilege** (Class 3) applies to devices too: a technician who only reads the switch status does not need administrator rights.
- Larger networks centralize device logins on an **AAA** (Authentication, Authorization, and Accounting) server (**RADIUS** (Remote Authentication Dial-In User Service) or **TACACS+** (Terminal Access Controller Access-Control System Plus)): accounts are created and removed in one place (**CIS** (Center for Internet Security) Safeguard 12.5).



With what record?

- Log successful and failed logins.
- Set an idle session timeout and limit the number of failed attempts.
- Show a **login banner**: access is restricted to authorized persons. It informs users and supports legal action against intruders.



Anti-lockout rule. pfSense and OPNsense always allow **GUI** (Graphical User Interface) access from the **LAN** (Local Area Network), so a mistaken rule cannot lock you out. Keep it until your own, more restrictive rule is tested, and document the decision in your manual.

Disabling what is not needed

Hardening principle from Class 5: every enabled service adds to the attack surface. On network equipment, three usual candidates:



UPnP (Universal Plug and Play)

Lets any internal device ask the router to open inbound ports automatically, without an administrator's decision.

Disabled on company routers.



WPS (Wi-Fi Protected Setup) **on access points**

With the **PIN** (Personal Identification Number) method the device checks the eight-digit **PIN** in two halves: about 11 000 combinations. An attacker within radio range can try them all in a few hours.

Disabled completely.



Remote and unused features

Remote management from the **WAN** (Wide Area Network), unused cloud management portals, unused switch ports.

Switched off; unused ports shut down or placed in an unused VLAN (Virtual Local Area Network).

Firmware and physical access

Firmware: the software stored in a device's non-volatile memory that starts and runs it. On a router or an access point an update replaces the whole embedded operating system; on OPNsense it is a new release of the system and its packages.

Three rules

- 1 Record the installed version of every device (the LAB_05 device registry).
- 2 Obtain firmware only from the manufacturer's website or built-in update service, and compare the published **SHA-256** (Secure Hash Algorithm, 256-bit) checksum before installing a downloaded image.
- 3 Replace a device past its **end of life**: after that date the manufacturer stops publishing security updates.



Physical access

- A console port gives direct command-line access and often a password-recovery procedure.
- Keep the equipment in a locked room or cabinet.
- Protect the console with a password.
- Treat configuration backups as sensitive files: they can contain password hashes, keys and a complete description of the network.

Wireless equipment: two secrets, one management rule



Administrator password of the access point

Opens the device's management plane. The login is reachable only from the wired side of the network.



Wi-Fi passphrase

Typed by users to join the wireless network. It protects the radio link, not the device.

The two secrets must be different.

Class 13: you add a guest **SSID** (Service Set Identifier) in its own zone and compare **WPA2** (Wi-Fi Protected Access 2) and **WPA3** (Wi-Fi Protected Access 3).

Tonight: the access point is managed exactly like the switch and the firewall (checks C1 to C7).

The NorthMaple management-plane baseline (C1 to C7)

Table 1: used in Example 1 and in LAB_05

Check	Requirement	Why it matters	Evidence for LAB_05 and Chapter 5
C1	Default password changed; named administrator account in use	Factory credentials are public and identical on many units (CIS (Center for Internet Security) Safeguard 4.7)	Screenshot of the user list showing the named account (laboratory-only values)
C2	Web GUI (Graphical User Interface) over HTTPS (Hypertext Transfer Protocol Secure) only (HTTP (Hypertext Transfer Protocol) disabled or redirected)	HTTP sends the login password in cleartext	Screenshot of the web GUI protocol setting; Example 2 Part B shows TCP (Transmission Control Protocol) 80 closed
C3	Command line over SSH (Secure Shell) only, or disabled; Telnet disabled	Telnet sends every keystroke in cleartext	Screenshot of the SSH setting (enabled only if needed)
C4	Management reachable only from the approved source NM-LWS01	Limits who can reach the login page at all	Screenshot of the rule or listen-interface setting
C5	SNMP (Simple Network Management Protocol) disabled, or SNMPv3 (Simple Network Management Protocol version 3) with authentication and encryption	SNMPv1 (Simple Network Management Protocol version 1) and SNMPv2c (Simple Network Management Protocol version 2c) send the community string in cleartext	Screenshot of the SNMP service status
C6	UPnP (Universal Plug and Play), WPS (Wi-Fi Protected Setup) and similar convenience features disabled	Each enabled service adds to the attack surface	Screenshot of the disabled services
C7	Firmware at the latest release; version recorded in the registry	Only a current release contains the published corrections	Screenshot of the version page; registry row



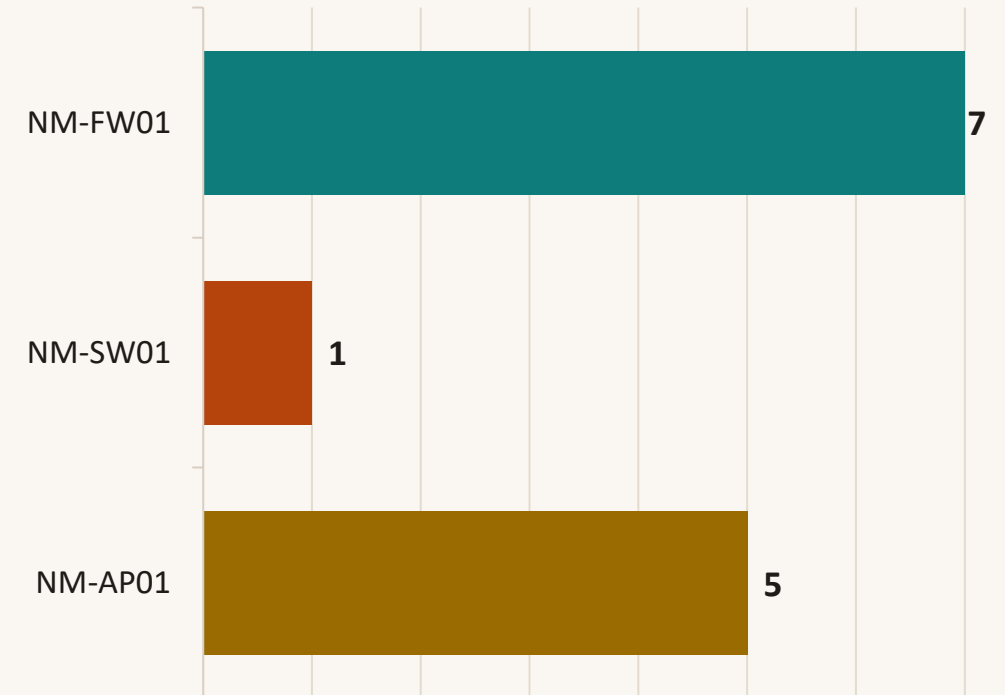
Example 1: the first-login audit of three devices

Code 1: `L1_device_mgmt_audit.py` (Python 3)

Task. Export the management settings of NM-FW01 (firewall), NM-SW01 (core switch) and NM-AP01 (access point), and check each one against C1 to C7.

- NM-SW01 is still close to its factory settings: default password, **HTTP** (Hypertext Transfer Protocol) and Telnet, management from anywhere, **SNMPv2c** (Simple Network Management Protocol version 2c), old firmware.
- NM-AP01 fails C4 (the guest Wi-Fi reaches the login page) and C6 (**WPS** (Wi-Fi Protected Setup) still enabled).
- The program saves the result as dated **JSON** (JavaScript Object Notation) evidence for the manual (Chapter 5, section 5.4).

Checks passed, out of 7



Feeds: LAB_05 part 1 · Final Project Chapter 5 · Manual section 5.4 · Class 13 guest Wi-Fi · resilience row 9



Lesson 1: watch and remember



Hardening Targets (CompTIA Security+ SY0-701, section 4.1)

Professor Messer · YouTube · under 15 minutes



Why this video

Covers default passwords, disabling unnecessary services, firmware updates and the hardening of network infrastructure devices, in the same order as sections 1.3 to 1.6.

Scan the code with your phone, or open:

<https://www.youtube.com/watch?v=YQKbs0ug0XQ>

Check the link with VirusTotal before opening it.



Five key terms to remember

Management plane

The services that configure and monitor a device: web **GUI** (Graphical User Interface), **CLI** (Command-Line Interface), **SNMP** (Simple Network Management Protocol) and **APIs** (Application Programming Interfaces).

Default credential

A factory username and password, public and identical on many units.

Management access restriction

Management services accept connections only from approved sources.

Anti-lockout rule

An automatic **LAN** (Local Area Network) rule that keeps **GUI** access open; keep it until your own rule is tested.

Firmware and end of life

Firmware runs the device; after the end-of-life date no security updates are published.

Why cleartext is not acceptable

- Between the workstation and the firewall, packets cross cables, switch ports and sometimes a wireless link.
- Anyone who can capture packets on that path can read them with a free packet analyzer such as Wireshark.
- A **cleartext protocol** sends its data without encryption: the username, the password and the content travel as readable characters.
- Cleartext protocols: Telnet, **HTTP** (Hypertext Transfer Protocol), **FTP** (File Transfer Protocol), **SNMPv1** (Simple Network Management Protocol version 1), **SNMPv2c** (Simple Network Management Protocol version 2c), and **IMAP** (Internet Message Access Protocol) without **TLS** (Transport Layer Security).

What a capture inside the laboratory shows

Telnet session (port 23): readable

```
login: admin
Password: Lab-Only-1
```

SSH session (port 22): ciphertext

```
5c 9e 01 d4 7a f3 2b 88
e0 41 c7 19 6d 0b a2 f5 ...
```

Laboratory-only values.



What a secure protocol provides



Confidentiality

Only the two endpoints can read the content, because it is encrypted.



Integrity

Any modification in transit is detected: a message authentication code, or an **AEAD** (Authenticated Encryption with Associated Data) cipher that encrypts and checks in one operation.



Authentication

Each side can verify the identity of the other: a certificate, a host key, a **PSK** (Pre-Shared Key) or a password.

The **handshake**: a short exchange at the start of every connection

1. Agree on the algorithms



2. Authenticate each other



3. Derive shared session keys
(for example Diffie-Hellman)



4. Protect the data with fast symmetric encryption (**AES** (Advanced Encryption Standard), ChaCha20)

Classes 14 and 15 study these algorithms and certificates in detail.

SSH (1/2)

Encrypted remote administration

- **SSH** (Secure Shell) is a protocol for encrypted remote login, remote command execution and file transfer.
- The **IETF** (Internet Engineering Task Force) defines its architecture in **RFC** (Request for Comments) 4251 and its transport layer in **RFC** 4253.
- **SSH** servers listen on **TCP** (Transmission Control Protocol) port 22 by default.
- **SFTP** (Secure Shell File Transfer Protocol) and **SCP** (Secure Copy Protocol) transfer files inside an **SSH** connection.

Port 22

the default port of an **SSH** server

Telnet · rlogin · rsh

sent everything in cleartext



SSH, protocol version 2

encrypted and authenticated



SSH (2/2)

How a connection proceeds

- 1 The client opens a **TCP** (Transmission Control Protocol) connection to port 22; both sides exchange version strings. Only protocol version 2 is acceptable.
- 2 Client and server negotiate the algorithms: key exchange, host-key type, cipher and integrity algorithm.
- 3 They perform the key exchange and derive the session keys. From this point, all traffic is encrypted.
- 4 The server proves its identity with its **host key**. First connection: compare the **fingerprint** with the server's console; it is then stored in `known_hosts`.
- 5 The user authenticates with a password or, preferably, **public-key authentication**: the private key never leaves the workstation.
- 6 The encrypted channel carries the shell session, the commands or the file transfers.



At NorthMaple: the administrator shell on the Linux file server, on the firewall when needed, on the switch when supported. Hardening: key-based authentication, no direct root login, reachable only from NM-LWS01. A host key that changes without explanation must be investigated.

SSL and TLS (1/2)

Which versions are allowed

- **TLS** (Transport Layer Security) protects most application traffic on the Internet. It runs on top of **TCP** (Transmission Control Protocol) and below the application protocol, so web, e-mail and directory protocols can all use it.
- **SSL** (Secure Sockets Layer) is its predecessor, designed by Netscape in the 1990s. People still say "**SSL** certificate", but every **SSL** version is now prohibited.



The versions to use: TLS 1.2 and TLS 1.3.

SSL and TLS (2/2)

The TLS 1.3 handshake, step by step

1. ClientHello

Supported **TLS** (Transport Layer Security) versions, cipher suites and a key share.



2. ServerHello

Chosen version and cipher suite, plus its own key share. Both sides compute the keys: the rest is encrypted.



3. Certificate, Verify, Finished

The **certificate** binds a public key to a name; CertificateVerify proves the server holds the private key.



4. Client checks, Finished

Signed by a **CA** (Certificate Authority) the client trusts? Valid today? Does the name match the server?



5. Application data

Encrypted and integrity-protected: for example the **GUI** (Graphical User Interface) login page and the password.



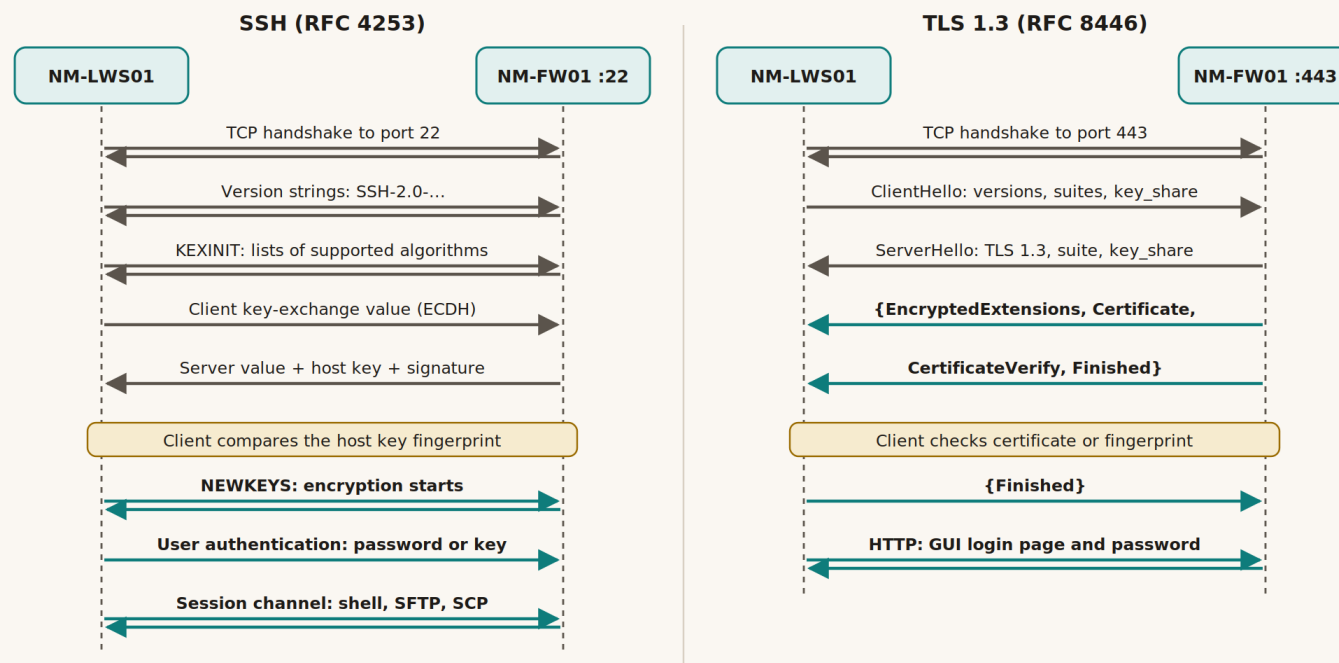
Cipher suite: the set of algorithms protecting the connection. Example 2 reads TLS_AES_256_GCM_SHA384 from the laboratory firewall: **AES** (Advanced Encryption Standard) with a 256-bit key in **GCM** (Galois/Counter Mode) mode, and **SHA-384** (Secure Hash Algorithm, 384-bit) for key derivation.

SSH and TLS 1.3 handshakes, side by side

Figure 2b

Figure 2b. SSH and TLS 1.3 handshakes between NM-LWS01 and NM-FW01

Teal arrows travel encrypted. Both protocols agree on keys before any password or command is sent.



TLS 1.3 needs one round trip before application data; the server's certificate is already encrypted. The first SSH connection shows a fingerprint: compare it with the value recorded when the appliance was installed, then keep it. A changed fingerprint later means stop and investigate.

Abbreviations: SSH (Secure Shell); TLS (Transport Layer Security); TCP (Transmission Control Protocol); KEXINIT (key exchange initialization); ECDH (Elliptic-Curve Diffie-Hellman); SFTP (SSH File Transfer Protocol); SCP (Secure Copy Protocol); HTTP (Hypertext Transfer Protocol); GUI (Graphical User Interface); RFC (Request for Comments).

1 round trip

TLS (Transport Layer Security) 1.3 before application data; TLS 1.2 needs two.



Self-signed certificate

- Signed with its own private key, not by a trusted **CA** (Certificate Authority): the browser shows a warning.
- pfSense and OPNsense create one for the web **GUI** (Graphical User Interface) at installation.
- In the laboratory: verify its fingerprint once and record it.

HTTPS: the web inside TLS

HTTP · port 80
cleartext

HTTPS · port 443
HTTP inside a TLS connection

- **HTTPS** (Hypertext Transfer Protocol Secure) is **HTTP** (Hypertext Transfer Protocol) carried inside **TLS** (Transport Layer Security), on **TCP** (Transmission Control Protocol) port 443.
- At NorthMaple it protects two flows: visitors reaching the public website in the **DMZ** (Demilitarized Zone), and the administrator reaching the firewall's web **GUI** (Graphical User Interface).
- For management, **HTTP** is disabled or reduced to an automatic redirect to **HTTPS**.
- The **HSTS** (Hypertext Transfer Protocol Strict Transport Security) header (**RFC** (Request for Comments) 6797) tells browsers to use only **HTTPS** for that site in the future.



Encryption is not access control

HTTPS protects the traffic between the browser and the server; it does not decide who is allowed to connect.

An **HTTPS** login page reachable from the Internet is still an exposed management plane.

Encryption and access restriction are two separate layers of defense-in-depth. NorthMaple needs both.

Layer 1: access restriction (C4): who may connect

Layer 2: encryption (C2, C3): nobody on the path can read it



IPsec (1/2)

Protection at the network layer

IPsec (Internet Protocol Security) is a suite of protocols that protects traffic at the network layer, the layer of **IP** (Internet Protocol) packets. Working below **TCP** (Transmission Control Protocol) and **UDP** (User Datagram Protocol), it protects every application at once, without changing them (architecture: **RFC** (Request for Comments) 4301).



IKE (Internet Key Exchange)

Today **IKEv2** (Internet Key Exchange version 2) (**RFC 7296**). Authenticates the two peers and negotiates the algorithms and keys.

UDP 500; UDP 4500 through a NAT
(Network Address Translation) **device (NAT-T** (Network Address Translation Traversal))



ESP (Encapsulating Security Payload)

RFC 4303. Encrypts the packets and protects their integrity.

IP protocol 50: not a TCP or UDP port



AH (Authentication Header)

Integrity and authentication without encryption. Does not work through **NAT**; rarely used today.

IP protocol 51

IPsec (2/2)

Security associations and the two modes

SA (Security Association): the agreement between two peers on the algorithms, the keys and their lifetime.

Tunnel mode

- The entire original packet is encrypted and placed inside a new packet with a new **IP** (Internet Protocol) header.
- Used between two firewalls to connect two offices: a **site-to-site VPN** (Virtual Private Network).

New IP header

ESP

Encrypted: original IP header + data

Transport mode

- Only the payload is protected; the original **IP** header is kept.
- Used between two hosts that communicate directly.

Original IP header

ESP

Encrypted: data

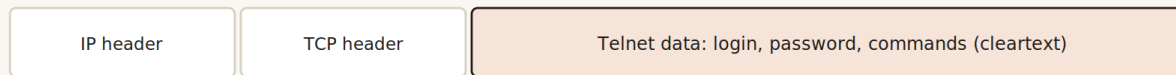
What each protocol encrypts

Figure 2a

Figure 2a. Which part of the packet each protocol protects

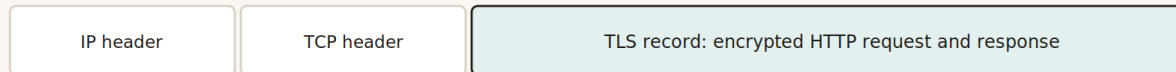
Shaded = encrypted and integrity-protected. White = readable by anyone who captures the traffic.

Telnet to NM-SW01 (TCP 23)



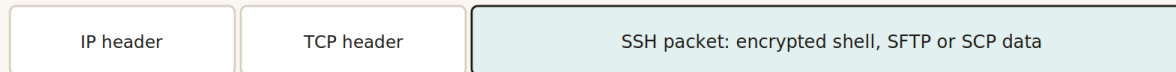
Everything readable, including the password.

HTTPS to the NM-FW01 GUI (TCP 443)



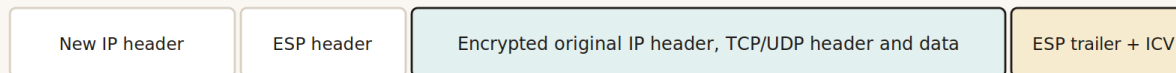
TLS protects one application connection.

SSH to NM-FS01 (TCP 22)



SSH protects its own sessions.

IPsec tunnel, ESP (IP protocol 50)



IPsec protects every IP packet between two gateways.

In tunnel mode the original packet, header included, is encrypted inside a new packet. The ICV lets the receiver detect any modification. LAB_06 builds the remote-access VPN with WireGuard (UDP 51820 by default), which applies the same idea with its own protocol.

Abbreviations: IP (Internet Protocol); TCP (Transmission Control Protocol); UDP (User Datagram Protocol); HTTP(S) (Hypertext Transfer Protocol (Secure)); TLS (Transport Layer Security); SSH (Secure Shell); SFTP (SSH File Transfer Protocol); SCP (Secure Copy Protocol); IPsec (Internet Protocol Security); ESP (Encapsulating Security Payload); ICV (Integrity Check Value); VPN (Virtual Private Network); GUI (Graphical User Interface).



NorthMaple and VPNs (Virtual Private Networks)

- **IPsec** (Internet Protocol Security) could protect a tunnel to a future second office, or the remote worker's connection.
- pfSense and OPNsense both include **IPsec**.
- LAB_06 builds the remote-access **VPN** (Virtual Private Network) with **WireGuard**, a simpler **VPN** protocol over **UDP** (User Datagram Protocol).
- Same questions for both: keys, users, allowed networks.



Choosing the right protocol (1/2)

Table 2: replace each cleartext protocol and record the port the firewall must allow

Task	Cleartext protocol (port)	Secure replacement (port)	Where NorthMaple uses it
Remote command line	Telnet (TCP (Transmission Control Protocol) 23)	SSH (Secure Shell) (TCP 22)	Administrator shell on NM-FS01, on NM-FW01 when needed, and on NM-SW01 when supported
Web management and websites	HTTP (Hypertext Transfer Protocol) (TCP 80)	HTTPS (Hypertext Transfer Protocol Secure) (TCP 443)	Firewall web GUI (Graphical User Interface) (from NM-LWS01 only); public website on NM-WEB01 in the DMZ (Demilitarized Zone)
File transfer	FTP (File Transfer Protocol) (TCP 21 control, TCP 20 data)	SFTP (Secure Shell File Transfer Protocol) or SCP (Secure Copy Protocol) inside SSH (TCP 22)	Administrator copies configuration backups to NM-FS01
Device monitoring	SNMPv1 (Simple Network Management Protocol version 1) / SNMPv2c (Simple Network Management Protocol version 2c) (UDP (User Datagram Protocol) 161, traps UDP 162)	SNMPv3 (Simple Network Management Protocol version 3) with authentication and privacy (UDP 161)	Optional monitoring of NM-SW01; otherwise SNMP (Simple Network Management Protocol) stays disabled
Mailbox access	IMAP (Internet Message Access Protocol) (TCP 143)	IMAPS (Internet Message Access Protocol Secure) (TCP 993)	Staff mail clients
Protecting every flow between two sites	Unprotected IP (Internet Protocol) routing	IPsec (Internet Protocol Security): IKE (Internet Key Exchange) (UDP 500, UDP 4500 with NAT-T (Network Address Translation Traversal)), ESP (Encapsulating Security Payload) (IP protocol 50)	Possible tunnel to a future second office; remote access is built with WireGuard (UDP 51820 by default) in LAB_06



Choosing the right protocol (2/2)

Two limits to remember



Encrypted traffic is opaque to the IDS (Intrusion Detection System)

- The LAB_04 sensor sees addresses, ports and packet sizes, but not the content.
- Keeping management flows few and predictable makes an unusual flow easier to notice.



Weak settings give weak protection

- An old **TLS** (Transport Layer Security) version, an **SSH** (Secure Shell) server that accepts any password, or an **IPsec** (Internet Protocol Security) tunnel with a short **PSK** (Pre-Shared Key) reduces the protection the protocol was designed to give.

What the LAB_04 sensor still sees in an encrypted flow:

Addresses

Ports

Packet sizes

Content: hidden



Compare your server settings with the Mozilla **SSL** (Secure Sockets Layer) Configuration Generator:

<https://ssl-config.mozilla.org/>

Example 2: which protocols is the workstation speaking?

Code 2: `L2_protocol_check.sh` (Bash)

Task. On NM-LWS01: list and classify the listening services (Part A), test the firewall's management ports (Part B), read the **TLS** (Transport Layer Security) parameters of its web **GUI** (Graphical User Interface) (Part C).

Part A: listening services

tcp 22	SSH	ENCRYPTED
tcp 23	Telnet	CLEARTEXT
udp 68	unlisted	REVIEW
tcp 631	unlisted	REVIEW
udp 5353	unlisted	REVIEW

1 cleartext service (Telnet): to be removed.

Part B: management ports

- port 80 (HTTP): closed
- port 443 (HTTPS): open
- port 22 (SSH): open

Target state for checks C2, C3 and C4, from NM-LWS01 only.

Part C: the web GUI's TLS

Protocol version: TLSv1.3

Ciphersuite: TLS_AES_256_GCM_SHA384

Verification: self-signed certificate

Expected in the laboratory: compare the fingerprint on first connection.

Feeds: LAB_05 evidence for C2 and C3 · LAB_05 part 2 test style · Class 15 and LAB_06 (certificates, VPN ports)



Lesson 2: watch and remember



How Secure Shell Works (SSH)

Computerphile • YouTube • 9 min 20 s

<https://www.youtube.com/watch?v=ORcvSkgdA58>



SSL, TLS, HTTPS Explained

ByteByteGo • YouTube • About 6 minutes

<https://www.youtube.com/watch?v=j9QmMEWmcfo>



Secure Protocols (CompTIA Security+ SY0-701, section 4.5)

Professor Messer • YouTube • under 15 minutes

<https://www.youtube.com/watch?v=9NAKCyOtFH0>



Secure Communication (CompTIA Security+ SY0-701, section 3.2)

Professor Messer • YouTube • under 15 minutes

https://www.youtube.com/watch?v=uU3e_ntg-3g



Five key terms to remember

Confidentiality, integrity, authentication

Only the endpoints read the data; changes are detected; each side verifies the other.

SSH (Secure Shell)

Encrypted remote login, commands and file transfer on TCP (Transmission Control Protocol) port 22.

TLS (Transport Layer Security) 1.3

The current TLS version (RFC (Request for Comments) 8446): keys are agreed before any password travels.

HTTPS (Hypertext Transfer Protocol Secure)

HTTP (Hypertext Transfer Protocol) inside TLS on TCP port 443: encryption, not access control.

IPsec (Internet Protocol Security)

Network-layer protection with IKE (Internet Key Exchange) and ESP (Encapsulating Security Payload); tunnel mode links two sites.

Vulnerability vocabulary (1/2)

Weakness, identifier, severity

- A **vulnerability** is a weakness in software, firmware or configuration that an attacker can use to defeat a security property.
- A public vulnerability receives a **CVE** (Common Vulnerabilities and Exposures) identifier (CVE-year-number) through the **CVE** Program: everyone names the same flaw the same way.
- Its severity is a **CVSS** (Common Vulnerability Scoring System) score from 0.0 to 10.0, maintained by **FIRST** (Forum of Incident Response and Security Teams):

Low
0.1 to 3.9

Medium
4.0 to 6.9

High
7.0 to 8.9

Critical
9.0 to 10.0



NVD (National Vulnerability Database)

Run by **NIST** (National Institute of Standards and Technology).
Publishes the details and the score of every **CVE** entry.



KEV (Known Exploited Vulnerabilities) **catalog**

Published by **CISA** (Cybersecurity and Infrastructure Security Agency):
vulnerabilities for which exploitation has been observed. A **KEV**
entry means attackers already use the flaw.

Vulnerability vocabulary (2/2)

Patch, update, upgrade and the process that ties them together



Patch

A correction published by the vendor.



Update

Installs a newer release that contains patches.



Upgrade

Moves to a new major version.



Zero-day vulnerability: attackers exploit it before any patch exists.

Patch management: the process of

identifying



prioritizing



acquiring



testing



installing



verifying

updates across an organization. **NIST** (National Institute of Standards and Technology) **SP** (Special Publication) 800-40 Revision 4 calls it preventive maintenance for technology.

Why network devices need their own update procedure



1. It usually needs a reboot

While the firewall reboots, the whole office loses Internet access and every zone-to-zone flow stops.



2. A failed update can stop the device

The device may be unable to start, so a tested way back, a **rollback**, is required.



3. Only the vendor can correct it

You depend on the vendor's security advisories and release schedule.



OPNsense release rhythm: two major releases per year, in January and July, with minor updates in between. Current series: 26.7 (July 2026); minor update 26.7.4 published on September 15, 2026.

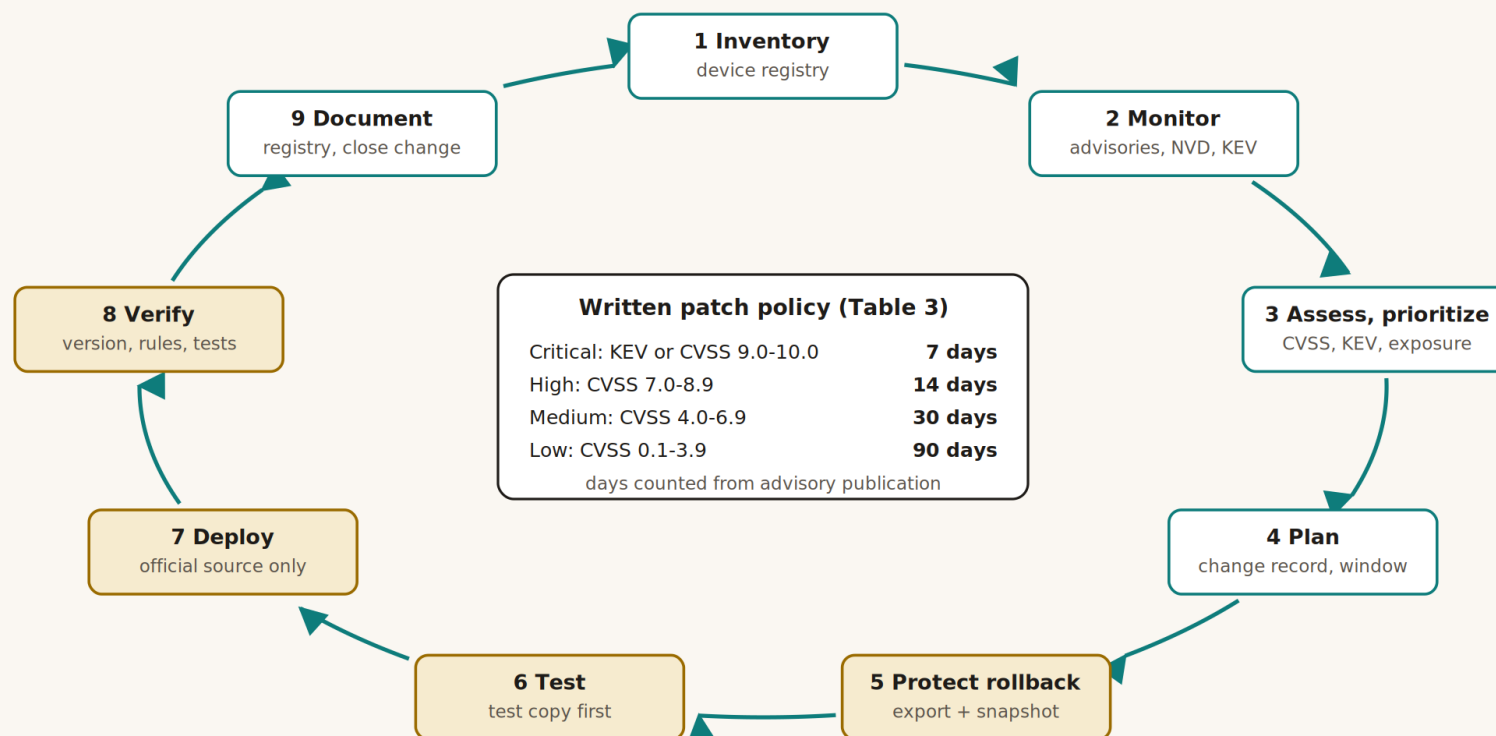
pfSense publishes its release notes on the Netgate documentation site.

The patch management process: nine steps

Figure 3a

Figure 3a. NorthMaple patch management cycle for network devices

Nine steps repeated for every advisory. Steps 5 to 8 happen inside one planned maintenance window.



Abbreviations: NVD (National Vulnerability Database); KEV (Known Exploited Vulnerabilities); CVSS (Common Vulnerability Scoring System).



Reading the cycle

- Steps 5 to 8 happen in one planned maintenance window, outside business hours.
- Monitor: vendor advisories, the **NVD** (National Vulnerability Database), the **CISA** (Cybersecurity and Infrastructure Security Agency) **KEV** (Known Exploited Vulnerabilities) catalog, the Canadian Centre for Cyber Security.
- Rollback: export the configuration; in the laboratory, take a **VM** (Virtual Machine) snapshot.
- Verify: version, interfaces, rules, **VPN** (Virtual Private Network) tunnels, logging.

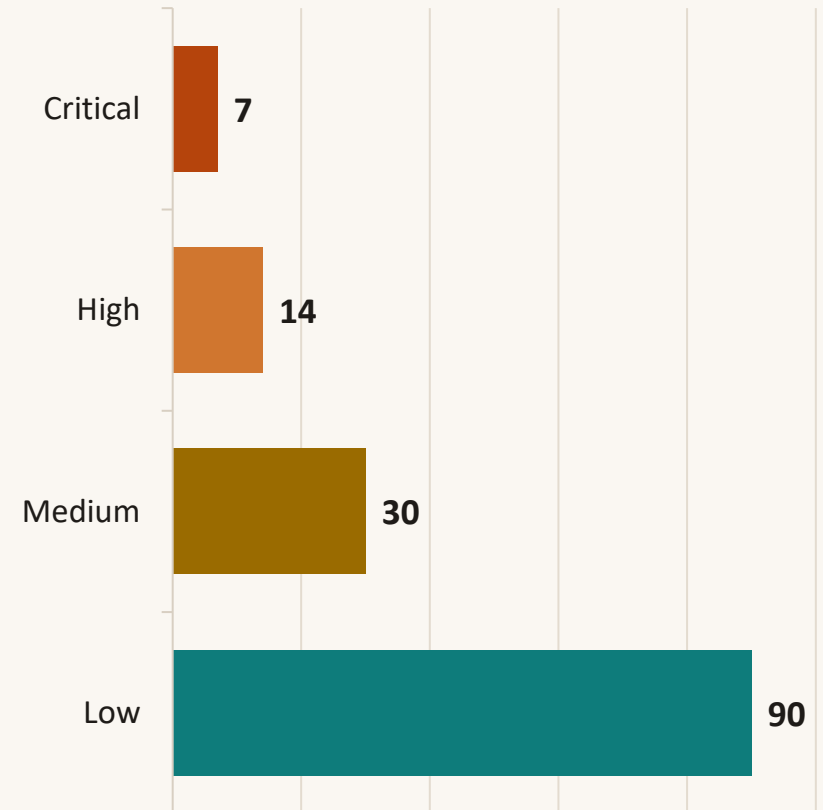
A written patch policy

Table 3: priorities are decided before an advisory arrives

Priority	Condition	Max. days	Typical action
Critical	Listed in the CISA (Cybersecurity and Infrastructure Security Agency) KEV (Known Exploited Vulnerabilities) catalog, or CVSS (Common Vulnerability Scoring System) 9.0 to 10.0	7	Emergency change; compensating control the same day if the patch must wait
High	CVSS 7.0 to 8.9	14	Next planned maintenance window
Medium	CVSS 4.0 to 6.9	30	Monthly maintenance window
Low	CVSS 0.1 to 3.9	90	Quarterly maintenance or next scheduled release

- Days count from the day the vendor publishes the advisory.
- Exposure can raise a priority: a High vulnerability in a **GUI** (Graphical User Interface) reachable from the Internet deserves Critical treatment.
- This is the example policy of a fictional company, not a legal requirement.

Maximum days from advisory publication



When no patch is available: compensating controls

Compensating controls reduce the risk until the vulnerability can be removed: when the vendor has not published a correction yet, or when the device is past its end of life.



Disable the vulnerable function



Restrict the affected service to the management source only



Add an IDS (Intrusion Detection System) rule that detects exploitation attempts



Plan the replacement of the device



Record the decision, its reason and its expiry date in the change record.

Measuring whether the patch process works

0

Target for overdue items. Example 3 finds 1 (NM-WEB01).

50 %

Devices on the latest release in Example 3 (2 of 4).

Time to patch

Days between the advisory and the verified installation of the correction.

When a number gets worse, the cause is usually one of three:

A device missing from the inventory

An advisory source nobody follows

A maintenance window cancelled without a new date



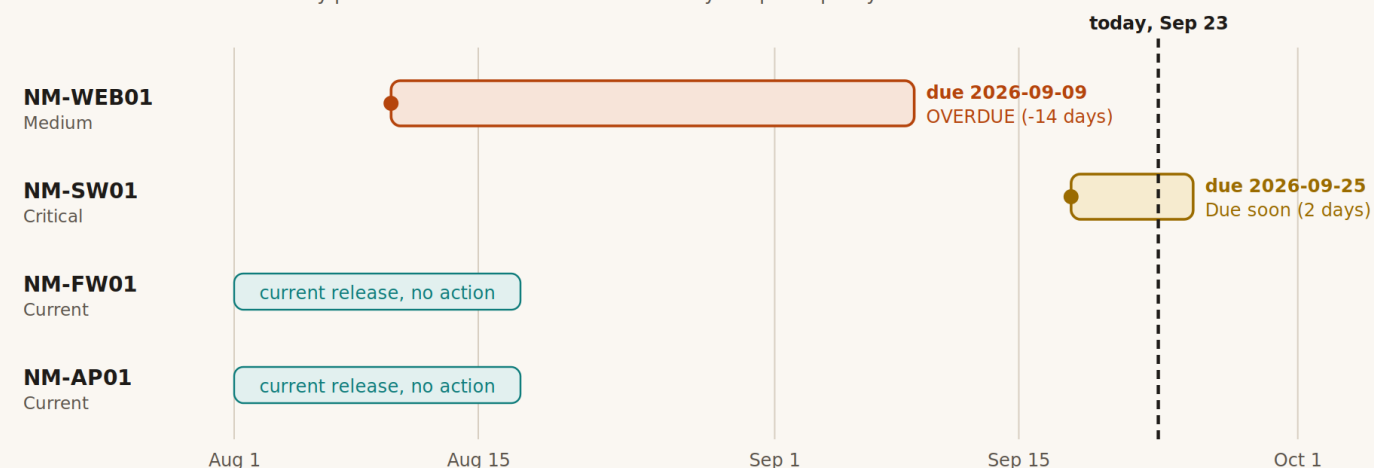
Laboratory rule. Download updates only from the appliance's official update mechanism or the vendor's website. The firewall must never scan or test any network other than your own laboratory.

Example 3: a vendor advisory arrives

Code 3: *L3_patch_plan.ps1* (PowerShell) · Figure 3b

Figure 3b. NorthMaple patch plan on Wednesday, September 23, 2026 (output of Example 3)

Each bar runs from the advisory publication date to the due date set by the patch policy.



Reading the plan: the Medium item on NM-WEB01 is 14 days late and is scheduled first; the KEV-listed switch firmware must be installed by September 25.

Abbreviation: KEV (Known Exploited Vulnerabilities). The advisories are fictional training data.



What the plan says

Task: apply Table 3 to four devices and print the plan, most urgent first.

- NM-WEB01: Medium, 14 days OVERDUE, scheduled first.
- NM-SW01: Critical (**KEV** (Known Exploited Vulnerabilities), **CVSS** (Common Vulnerability Scoring System) 9.8), due September 25.
- NM-FW01 and NM-AP01: current, no action.

Every update: export, snapshot, install, verify, document.

Feeds: the registry's Update status column (LAB_05, Chapter 5) · LAB_05 step 6, snapshot before any update · Manual section 5.5 · resilience rows 3 and 9



Lesson 3: watch and remember



Vulnerability Remediation (CompTIA Security+ SY0-701, section 4.3)

Professor Messer · YouTube · under 15 minutes



Why this video

Covers patching, compensating controls, exceptions and verification after remediation (sections 3.3 to 3.5).

Scan the code with your phone, or open:

<https://www.youtube.com/watch?v=P9xakfmX70c>

Check the link with VirusTotal before opening it.



Five key terms to remember

CVE (Common Vulnerabilities and Exposures)

The public identifier of one vulnerability, of the form CVE-year-number.

CVSS (Common Vulnerability Scoring System) **severity bands**

Low 0.1 to 3.9, Medium 4.0 to 6.9, High 7.0 to 8.9, Critical 9.0 to 10.0.

KEV (Known Exploited Vulnerabilities) **catalog**

The **CISA** (Cybersecurity and Infrastructure Security Agency) list of vulnerabilities that attackers already exploit.

Patch management process

Inventory, monitor, assess, plan, protect the rollback, test, deploy, verify, document.

Compensating control

A measure that reduces the risk until the vulnerability can be removed.

An inventory comes first



CIS (Center for Internet Security) Control 1: Inventory and Control of Enterprise Assets

The first of the eighteen **CIS** Controls. Safeguard 1.1 asks for a detailed enterprise asset inventory.

- **Asset inventory**: the list of the hardware and software an organization owns.
- **Device registry**: the part of the inventory that covers network devices and appliances, with the details needed to operate them securely.

A device missing from the inventory is:



not updated



not backed up



not included in the firewall rules

What goes in the device registry

Required by Final Project Chapter 5, for every network device and appliance

Role

Model or VM (Virtual Machine)

Software version

Management access method

Update status

Added for NorthMaple

Unique device name (matches the Chapter 1 diagram)

Zone

Management IP (Internet Protocol) address

Owner (responsible person)

Last configuration change

Last backup: date and location

End-of-life date, when known



Never passwords or keys

At most, the registry names the password-manager entry where they are stored.



A sensitive document

It describes the network and its weak points: access is limited to the IT (Information Technology) staff.

Baseline, drift and change management



Configuration baseline

An approved, documented set of settings: the reference a device is checked against. Checks C1 to C7 are part of NorthMaple's baseline.



Configuration drift

The difference that appears over time between the baseline and the running configuration: undocumented changes, emergency fixes, forgotten tests.



Change management

Every change is requested, reviewed for risk, approved, performed in a planned window with a rollback plan, verified and documented.

NIST (National Institute of Standards and Technology) **SP** (Special Publication) **800-128: security-focused configuration management in four phases**

1. Planning



2. Identifying and implementing configurations



3. Controlling configuration changes



4. Monitoring

For a ten-person accounting firm the result fits on one page: the configuration management plan.



NorthMaple's one-page configuration management plan

Table 4

Plan element	NorthMaple decision
Scope	Every network device and appliance: NM-FW01, NM-SW01, NM-AP01, NM-IDS01; servers are added as the labs progress
Roles	The IT (Information Technology) administrator is the configuration owner and performs changes; the managing partner approves any change that interrupts service
Baselines	Management-plane baseline C1 to C7 (Lesson 1); zone-to-zone firewall rules (LAB_05 part 2); secure protocol choices (Table 2)
Change procedure	Request, risk review, approval, planned window outside business hours, configuration export and VM (Virtual Machine) snapshot, change, verification, documentation
Backups	Configuration exported before and after every change and at least every 30 days; encrypted; stored in the protected IT share on NM-FS01 plus one offline copy
Monitoring and drift	Monthly comparison of the running configuration with the last approved backup; weekly check of vendor advisories and updates
Records	Device registry, change log, backup log, patch plan
Review	Plan reviewed every six months and after any security incident

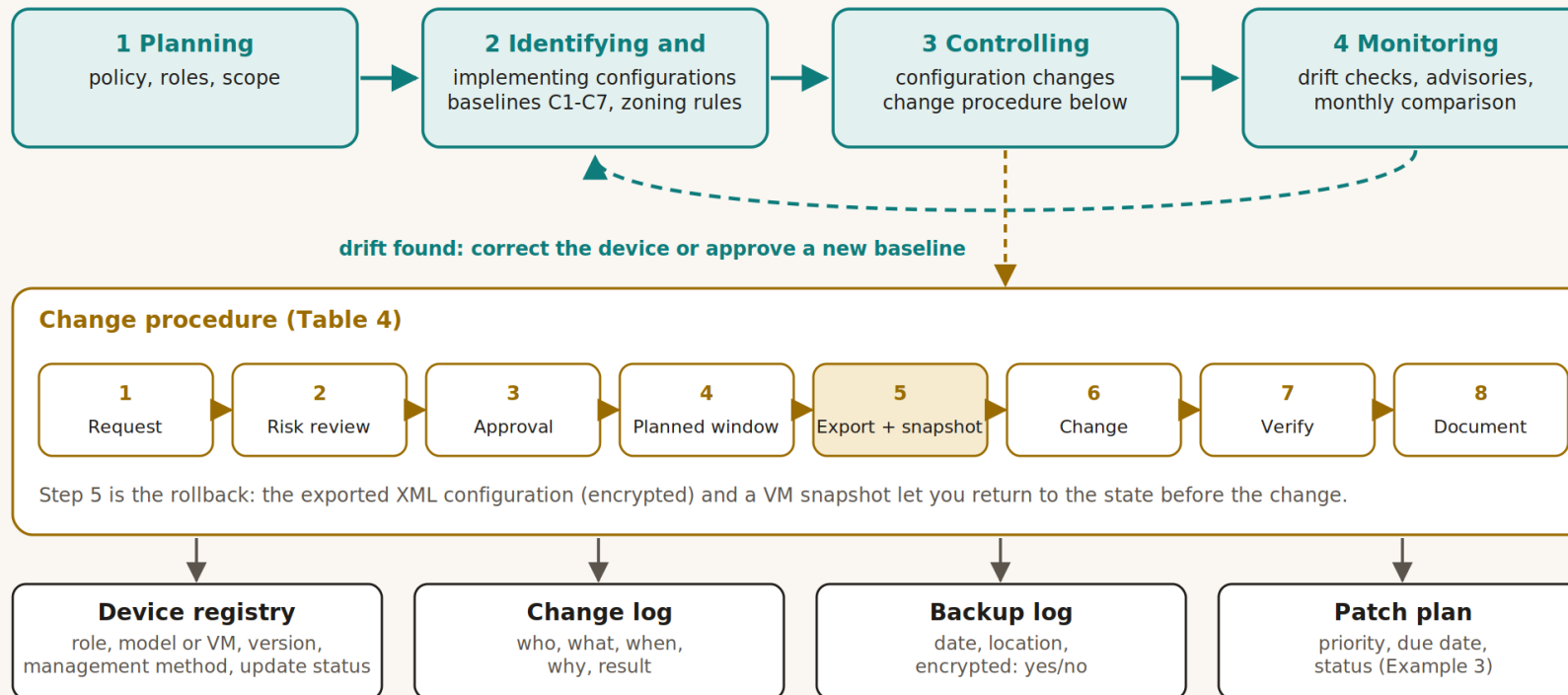


Configuration management at NorthMaple

Figure 4, after NIST SP 800-128

Figure 4. Security-focused configuration management at NorthMaple (after NIST SP 800-128)

Four phases on top; the change procedure that implements the third phase; the records that prove it.



Abbreviations: NIST (National Institute of Standards and Technology); SP (Special Publication); XML (Extensible Markup Language); VM (Virtual Machine).



The configuration file

One **XML** (Extensible Markup Language) file holds everything.

pfSense and OPNsense keep the whole configuration in one file, with previous versions.

The **GUI** (Graphical User Interface) can download, restore and compare two versions: this detects drift.

The file holds password hashes and keys: encrypt the backup.



How the four records work together



Device registry

What do we have, and in what state is it?



Change log

Who changed what, when and why?



Backup log

Can we return to a known good state, and from which file?



Patch plan

Which correction is due, and by when?

One event, four records: on September 25, firmware 3.2.0 is installed on NM-SW01

1. Export the configuration



2. Record it in the backup log



3. Install, verify the version and repeat the tests



4. Registry: version 3.2.0, status Current



5. Close the change record

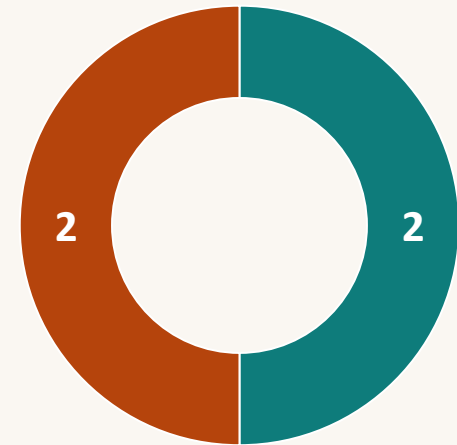
A skipped step shows up in the next monthly drift check. Final Project Chapter 5 is graded on this evidence: the registry, the rule set and the allowed/blocked tests.

Example 4: starting the company device registry

Code 4: L4_device_registry.html (self-contained page)

Device	Version	Management	Owner	Last change	Last backup
NM-FW01	26.7.4	HTTPS (Hypertext Transfer Protocol Secure), SSH (Secure Shell)	IT (Information Technology) administrator	2026-09-23	2026-09-23
NM-SW01	3.1.2	HTTP (Hypertext Transfer Protocol), Telnet	IT administrator	2026-09-10	2026-08-01
NM-AP01	5.4.1	HTTPS	IT administrator	2026-09-01	2026-09-01
NM-IDS01	(missing)	SSH	(missing)	2026-09-16	2026-09-16

Registry health on September 23: 2 of 4 devices (50 %)



■ Meets every rule ■ Needs work

- R1 mandatory fields filled (Owner)
- R2 no **HTTP** or Telnet for management
- R3 software version recorded
- R4 backup at least as recent as the last change
- R5 backup at most 30 days old

The registry never stores a password.



Lesson 4: watch and remember



Asset Management (CompTIA Security+ SY0-701, section 4.2)

Professor Messer · YouTube · under 15 minutes

Explains asset inventory, ownership and tracking, the basis of the device registry (sections 4.1 and 4.2).

<https://www.youtube.com/watch?v=BJ2UMB4a04g>



Technical Change Management (CompTIA Security+ SY0-701, section 1.3)

Professor Messer · YouTube · under 15 minutes

Covers allow and deny lists, restarts, dependencies and documentation of changes (section 4.3).

<https://www.youtube.com/watch?v=H9TYNjcpl-0>



Five key terms to remember

Device registry

The inventory of network devices, with the details needed to run them securely.

Configuration baseline

The approved, documented settings a device is checked against.

Configuration drift

The difference that appears over time between the baseline and the running configuration.

Change management

Every change is requested, reviewed, approved, planned, verified and documented.

NIST (National Institute of Standards and Technology) **SP** (Special Publication) **800-128 phases**

Planning; identifying and implementing configurations; controlling changes; monitoring.

LAB_05, part 1 (1/2)

What you will do after the break

Goal: place pfSense or OPNsense as a **VM** (Virtual Machine) in front of the NorthMaple network, secure its management plane, and start the company device registry.

1

Setup. Create the firewall **VM** in your isolated laboratory: one **NIC** (Network Interface Card) facing the laboratory's outside network (**WAN** (Wide Area Network)), one facing the company network (**LAN** (Local Area Network)).

2

Setup. Install the appliance and complete the initial setup.

3

Lesson 1, C1. Replace the default administrator credentials; create your named administrator account.

4

Lesson 2. Configure **HTTPS** (Hypertext Transfer Protocol Secure) management; disable **HTTP** (Hypertext Transfer Protocol) or reduce it to a redirect; record the **GUI** (Graphical User Interface) certificate fingerprint.

5

Lesson 1, C3 and C4. Enable **SSH** (Secure Shell) only if you need the shell, and only for the administrator's workstation.

6

Lesson 3. Record the installed version, check for updates, take a snapshot before any update.

7

Lesson 4. Start the registry: the firewall first, then the switch, the access point and the **IDS** (Intrusion Detection System) sensor.



LAB_05, part 1 (2/2)

Evidence, part 2 and the rules



Evidence to capture

- Credential change (laboratory-only values)
- **HTTPS** (Hypertext Transfer Protocol Secure) and **SSH** (Secure Shell) settings
- Version page
- Your device registry
- Optional: outputs of tonight's four scripts



Where it goes

- Final Project Chapter 5: firewall, device registry, tests
- Course Procedure Manual Chapter 5
- Resilience assessment row 9: default credentials on network devices



Part 2: Class 12, Saturday, September 26

- Translate the Class 2 zoning table into real firewall rules between zones.
- Validate every rule with an allowed/blocked connection test.



Work only inside your isolated laboratory (Legal Notice, section 3); the **WAN** (Wide Area Network) of your firewall connects only to your own laboratory or host network. **AI** (Artificial Intelligence) tools: section 5.6 of your manual chapter states the **LLM** (Large Language Model) and the exact prompts. **The model drafts. The analyst decides. The analyst's name goes on the record.**

Official links (1/4)

Firewall appliances, baselines and advisories

Group	Resource	Official address (check it with VirusTotal first)
Firewall appliances used in LAB_05	OPNsense project home page	https://opnsense.org/
	OPNsense documentation	https://docs.opnsense.org/
	pfSense project home page (Netgate)	https://www.pfsense.org/
	pfSense documentation (Netgate)	https://docs.netgate.com/pfsense/en/latest/
	Oracle VirtualBox	https://www.virtualbox.org/
Baselines and controls	CIS (Center for Internet Security) Critical Security Controls	https://www.cisecurity.org/controls
	CIS Benchmarks	https://www.cisecurity.org/cis-benchmarks
	NIST (National Institute of Standards and Technology) SP (Special Publication) 800-40 Rev. 4: enterprise patch management planning	https://csrc.nist.gov/pubs/sp/800/40/r4/final
	NIST SP 800-128: security-focused configuration management	https://csrc.nist.gov/pubs/sp/800/128/upd1/final
	MITRE ATT&CK (Adversarial Tactics, Techniques, and Common Knowledge) T1078.001, Valid Accounts: Default Accounts	https://attack.mitre.org/techniques/T1078/001/
Vulnerabilities and advisories	USENIX Security 2017, Understanding the Mirai Botnet	https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/antonakakis
	CVE (Common Vulnerabilities and Exposures) Program	https://www.cve.org/
	NIST National Vulnerability Database	https://nvd.nist.gov/



Official links (2/4)

Advisories and protocol standards

Group	Resource	Official address (check it with VirusTotal first)
Vulnerabilities and advisories	FIRST (Forum of Incident Response and Security Teams), Common Vulnerability Scoring System	https://www.first.org/cvss/
	CISA (Cybersecurity and Infrastructure Security Agency) Known Exploited Vulnerabilities catalog	https://www.cisa.gov/known-exploited-vulnerabilities-catalog
	Canadian Centre for Cyber Security	https://www.cyber.gc.ca/en
Protocol standards	RFC (Request for Comments) 4251, The SSH (Secure Shell) Protocol Architecture	https://www.rfc-editor.org/rfc/rfc4251
	RFC 4253, The SSH Transport Layer Protocol	https://www.rfc-editor.org/rfc/rfc4253
	RFC 8446, The TLS (Transport Layer Security) Protocol Version 1.3	https://www.rfc-editor.org/rfc/rfc8446
	RFC 8996, Deprecating TLS 1.0 and TLS 1.1	https://www.rfc-editor.org/rfc/rfc8996
	RFC 6176, Prohibiting SSL (Secure Sockets Layer) Version 2.0	https://www.rfc-editor.org/rfc/rfc6176
	RFC 7568, Deprecating SSL Version 3.0	https://www.rfc-editor.org/rfc/rfc7568
	RFC 9110, HTTP (Hypertext Transfer Protocol) Semantics	https://www.rfc-editor.org/rfc/rfc9110
	RFC 6797, HTTP Strict Transport Security	https://www.rfc-editor.org/rfc/rfc6797
	RFC 4301, Security Architecture for the Internet Protocol	https://www.rfc-editor.org/rfc/rfc4301

Official links (3/4)

Protocol standards and tools

Group	Resource	Official address (check it with VirusTotal first)
Protocol standards	RFC (Request for Comments) 4303, IP (Internet Protocol) Encapsulating Security Payload	https://www.rfc-editor.org/rfc/rfc4303
	RFC 7296, Internet Key Exchange Protocol Version 2	https://www.rfc-editor.org/rfc/rfc7296
	RFC 3414, User-based Security Model for SNMPv3 (Simple Network Management Protocol version 3)	https://www.rfc-editor.org/rfc/rfc3414
	IANA (Internet Assigned Numbers Authority) port number registry	https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml
Tools	OpenSSH	https://www.openssh.com/
	OpenSSL	https://www.openssl.org/
	Mozilla SSL (Secure Sockets Layer) Configuration Generator	https://ssl-config.mozilla.org/
	Wireshark	https://www.wireshark.org/
	WireGuard	https://www.wireguard.com/
	Python	https://www.python.org/
	GNU (GNU's Not Unix) Bash	https://www.gnu.org/software/bash/
	PowerShell documentation (Microsoft Learn)	https://learn.microsoft.com/powershell/



Official links (4/4)

Tools, link safety and legal framework

Group	Resource	Official address (check it with VirusTotal first)
Tools	Wi-Fi Alliance	https://www.wi-fi.org/
Videos	Professor Messer (video courses and notes)	https://www.professormesser.com/
Link safety	VirusTotal URL (Uniform Resource Locator) scanner	https://www.virustotal.com/gui/home/url
Legal framework	Criminal Code of Canada (R.S.C. 1985, c. C-46)	https://laws-lois.justice.gc.ca/eng/acts/C-46/
	Québec private-sector privacy act (Law 25)	https://www.legisquebec.gouv.qc.ca/en/document/cs/P-39.1
	Commission d'accès à l'information du Québec	https://www.cai.gouv.qc.ca/
	PIPEDA (Personal Information Protection and Electronic Documents Act) (S.C. 2000, c. 5)	https://laws-lois.justice.gc.ca/eng/acts/P-8.6/
	Office of the Privacy Commissioner of Canada	https://www.priv.gc.ca/en/
	GDPR (General Data Protection Regulation), Regulation (EU (European Union)) 2016/679	https://eur-lex.europa.eu/eli/reg/2016/679/oj
	Directive 2013/40/ EU (attacks against information systems)	https://eur-lex.europa.eu/eli/dir/2013/40/oj
	US (United States) Computer Fraud and Abuse Act, 18 U.S.C. § 1030	https://www.law.cornell.edu/uscode/text/18/1030
	Spain, Ley Orgánica 10/1995 del Código Penal	https://www.boe.es/buscar/act.php?id=BOE-A-1995-25444

Legal notice and acceptable use

- **Educational use only**, inside your own isolated laboratory, or on a system whose owner gave you explicit written permission before you start.
- Unauthorized computer access or use is a criminal offence: Criminal Code of Canada, s. 342.1, s. 342.2 and s. 430(1.1).
- Personal information: Québec Law 25 and **PIPEDA** (Personal Information Protection and Electronic Documents Act). Outside Canada, for example: **GDPR** (General Data Protection Regulation), **US** (United States) **CFAA** (Computer Fraud and Abuse Act) (18 U.S.C. § 1030), **EU** (European Union) Directive 2013/40/**EU**, Spain's Ley Orgánica 10/1995.
- If an activity seems to touch a system you do not own, stop and ask the instructor first.
- **AI** (Artificial Intelligence) tools: state the **LLM** (Large Language Model) used and the exact prompts in your submission.



The model drafts. The analyst decides. The analyst's name goes on the record.

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